Week 5, Lecture 1
Quantum optical channels

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Office hours: Thursdays 3-4 pm

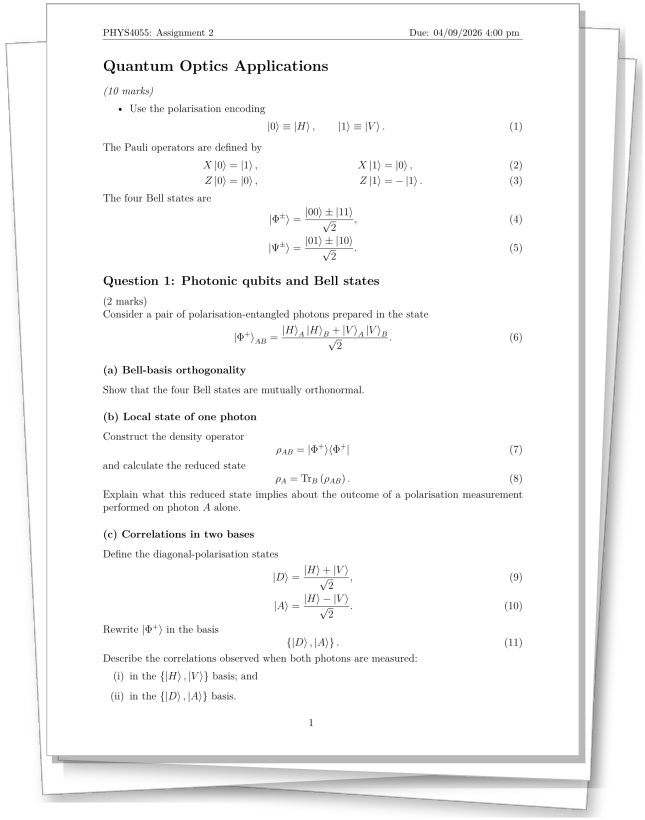
Bachor & Ralph, A Guide to Experiments in Quantum Optics, 3rd Edition
Walls & Milburn, Quantum Optics, 2nd Edition
Nielsen & Chuang, Quantum Computation and Quantum Information

Announcement

Assignment 2 is on black board in assessment folder

Assignment 2 covers quantum communication, all relevant content was covered in Week 4

Due **04/09/26 4pm** per course profile



PHYS4055: Assignment 2 Due: 04/09/2026 4:00 pm

Quantum Optics Applications

(10 marks)

- Use the polarisation encoding

$$|0\rangle \equiv |H\rangle\,, \qquad |1\rangle \equiv |V\rangle\,. \tag{1}$$

The Pauli operators are defined by

$$X|0\rangle = |1\rangle\,, \qquad X|1\rangle = |0\rangle\,, \tag{2}$$
$$Z|0\rangle = |0\rangle\,, \qquad Z|1\rangle = -|1\rangle\,. \tag{3}$$

The four Bell states are

$$|\Phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \tag{4}$$
$$|\Psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{5}$$

Question 1: Photonic qubits and Bell states

(2 marks)
Consider a pair of polarisation-entangled photons prepared in the state

$$|\Phi^{+}\rangle_{AB} = \frac{|H\rangle_A|H\rangle_B + |V\rangle_A|V\rangle_B}{\sqrt{2}}. \tag{6}$$

(a) Bell-basis orthogonality

Show that the four Bell states are mutually orthonormal.

(b) Local state of one photon

Construct the density operator

$$\rho_{AB} = |\Phi^{+}\rangle\langle\Phi^{+}| \tag{7}$$

and calculate the reduced state

$$\rho_A = \mathrm{Tr}_B\left(\rho_{AB}\right). \tag{8}$$

Explain what this reduced state implies about the outcome of a polarisation measurement performed on photon A alone.

(c) Correlations in two bases

Define the diagonal-polarisation states

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \tag{9}$$
$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \tag{10}$$

Rewrite $|\Phi^{+}\rangle$ in the basis

$$\{|D\rangle, |A\rangle\}. \tag{11}$$

Describe the correlations observed when both photons are measured:

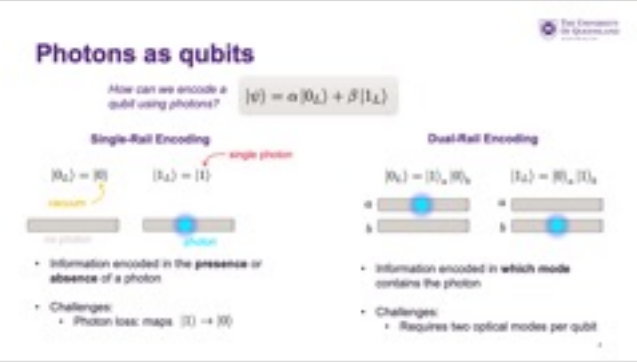
(i) in the $\{|H\rangle, |V\rangle\}$ basis; and

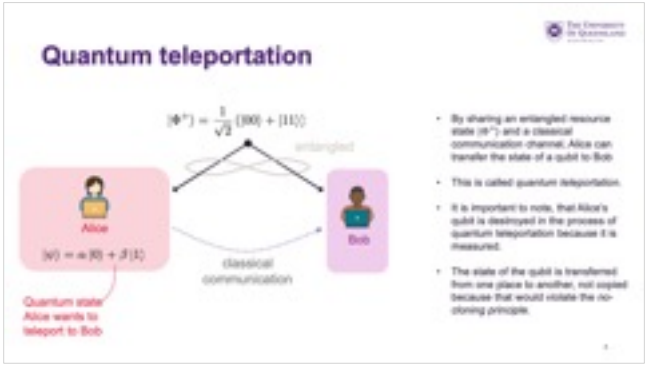
(ii) in the $\{|D\rangle, |A\rangle\}$ basis.

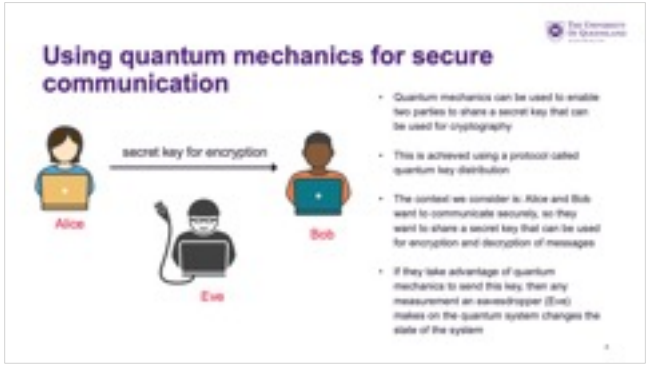
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Review

What you learned last week…





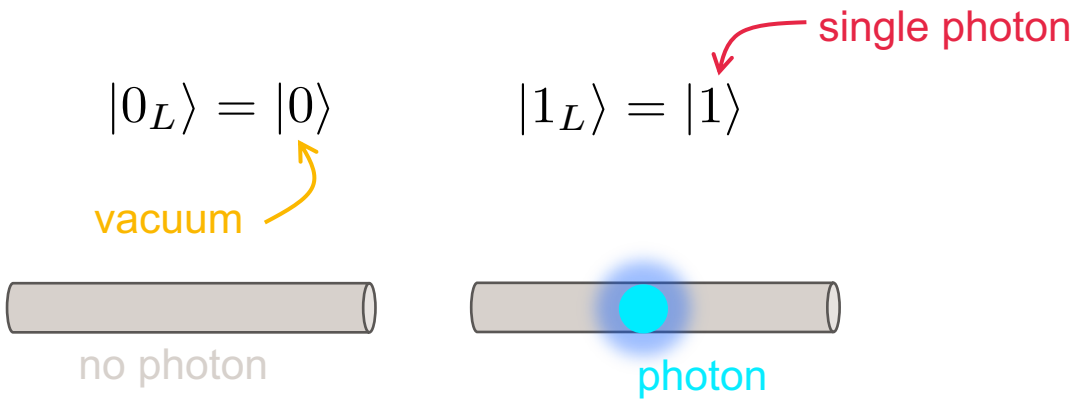


Photons as qubits

How can we encode a qubit using photons?

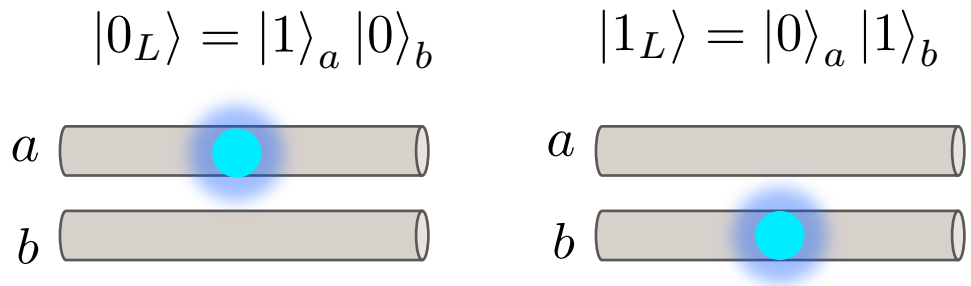
$$|\psi\rangle = \alpha\,|0_L\rangle + \beta\,|1_L\rangle$$

Single-Rail Encoding



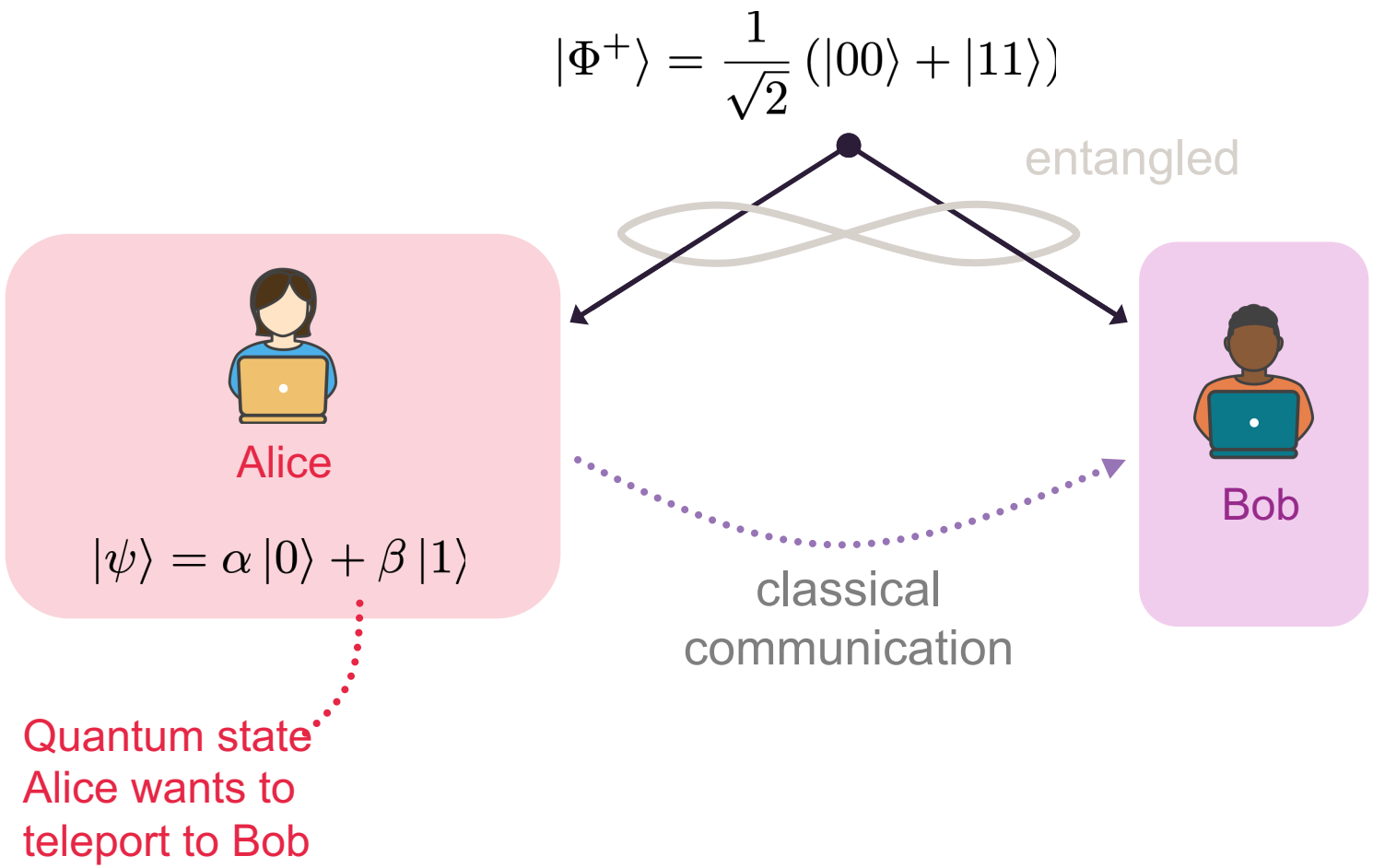
- Information encoded in the **presence** or **absence** of a photon
- Challenges:
  - Photon loss: maps $|1\rangle \rightarrow |0\rangle$

Dual-Rail Encoding



- Information encoded in **which mode** contains the photon
- Challenges:
  - Requires two optical modes per qubit

Quantum teleportation



- By sharing an entangled resource state $|\Phi^+\rangle$ and a classical communication channel, Alice can transfer the state of a qubit to Bob
- This is called *quantum teleportation.*
- It is important to note, that Alice's qubit is destroyed in the process of quantum teleportation because it is measured.
- The state of the qubit is transferred from one place to another, not copied because that would violate the *no-cloning principle.*

Using quantum mechanics for secure communication



- Quantum mechanics can be used to enable two parties to share a secret key that can be used for cryptography
- This is achieved using a protocol called quantum key distribution
- The context we consider is: Alice and Bob want to communicate securely, so they want to share a secret key that can be used for encryption and decryption of messages
- If they take advantage of quantum mechanics to send this key, then any measurement an eavesdropper (Eve) makes on the quantum system changes the state of the system

What you will learn today…

Why interactions with an environment produce mixed states and open-system dynamics.

How the Schrödinger and Heisenberg pictures describe optical quantum channels.

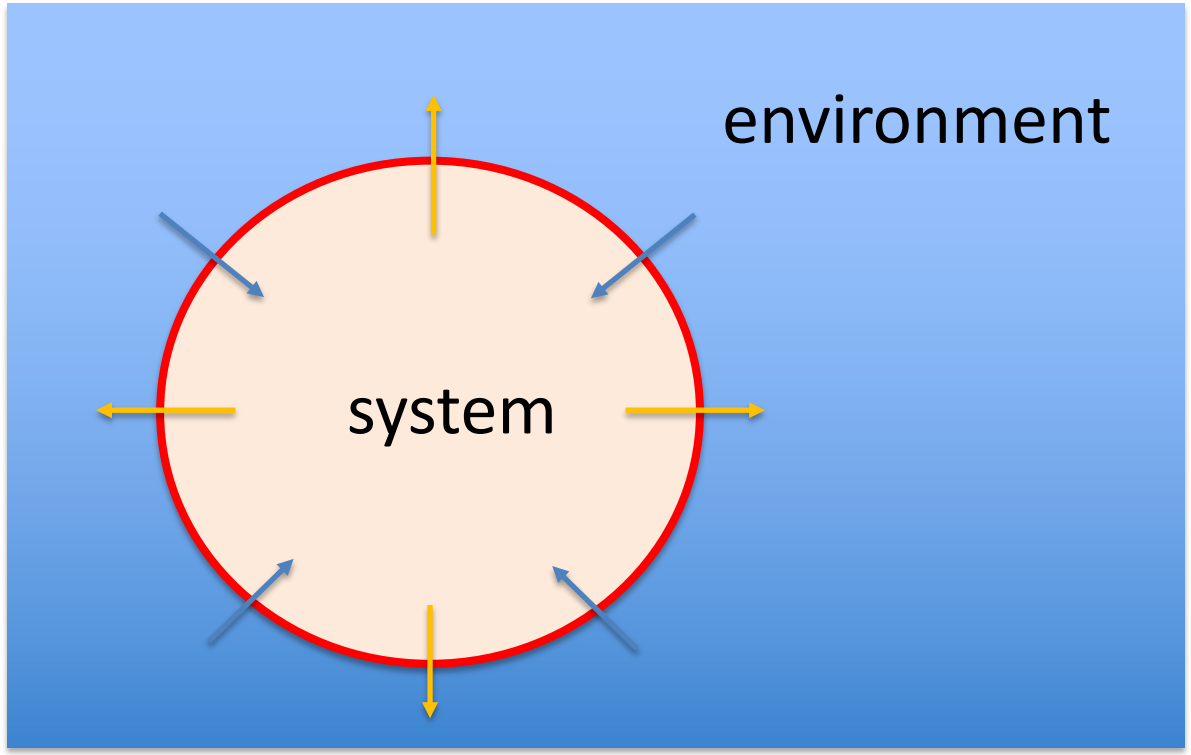
How amplitude damping and phase damping affect photon number and quantum coherence.

Open quantum optical systems

From unitary evolution to loss, noise and measurement

- An isolated quantum system undergoes unitary evolution.
- Real optical systems interact with an external environment.
- The combined system and environment evolve unitarily, but the reduced system generally does not:

$$\hat{\rho}_S(t) = \mathrm{Tr}_E \left[\hat{U}_{SE}(t)\, \hat{\rho}_{SE}(0)\, \hat{U}_{SE}^{\dagger}(t) \right]$$



Review: unitary evolution

For an isolated quantum system with a time-independent Hamiltonian $\widehat{H}$, the evolution operator is

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right)$$

The evolution is unitary:

$$\hat{U}^\dagger(t)\hat{U}(t) = \hat{U}(t)\hat{U}^\dagger(t) = \mathbb{I}$$

Schrödinger picture — states evolve

$$|\psi(t)\rangle = \hat{U}(t)\,|\psi(0)\rangle$$

Heisenberg picture — operators evolve

$$\hat{A}(t) = \hat{U}^\dagger(t)\hat{A}(0)\hat{U}(t)$$

Same physics, different pictures

Where does the time dependence live?

Schrödinger picture

The state evolves; observables remain fixed.

$$|\psi_S(t)\rangle = \hat{U}(t)\,|\psi(0)\rangle\,,$$
$$\hat{\rho}_S(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}^\dagger(t).$$

The density operator obeys

$$\frac{d\hat{\rho}_S(t)}{dt} = -\frac{i}{\hbar}\left[\hat{H}, \hat{\rho}_S(t)\right]$$

Heisenberg picture

The state remains fixed; observables evolve.

$$|\psi_H\rangle = |\psi(0)\rangle\,,$$
$$\hat{\rho}_H = \hat{\rho}(0),$$
$$\hat{A}_H(t) = \hat{U}^\dagger(t)\hat{A}_S\hat{U}(t).$$

For an observable with no explicit time dependence,

$$\frac{d\hat{A}_H(t)}{dt} = \frac{i}{\hbar}\left[\hat{H}, \hat{A}_H(t)\right]$$

Identical predictions: $\left\langle \hat{A} \right\rangle_t = \mathrm{Tr}\left[\hat{\rho}_S(t)\hat{A}_S\right] = \mathrm{Tr}\left[\hat{\rho}_H\hat{A}_H(t)\right]$

Unitary interactions in quantum optics

An optical mode $\hat{a}$ can interact with another quantum system through an operator $\hat{c}$:

$$\hat{H}_i = i\hbar\kappa\left(\hat{a}^\dagger\hat{c} - \hat{a}\hat{c}^\dagger\right)$$

where κ is the interaction strength.

Different choices of $\hat{c}$ describe different interactions:

$\hat{c} = \hat{b}$ beam-splitter interaction between two optical modes

$\hat{c} = \hat{\sigma}_-$ atom–field excitation exchange, equivalent to the Jaynes–Cummings interaction up to a phase convention

$\hat{c} = \hat{\sigma}_z$ coupling between a field quadrature and an atomic population, producing dephasing

For the beam-splitter interaction, the dimensionless mixing angle used previously is $\theta = \kappa t$

What does the operator $\hat{c}$ represent?

The choice of $\hat{c}$ determines the physical interaction described by $\hat{H}_i = i\hbar\kappa\left(\hat{a}^\dagger\hat{c} - \hat{a}\hat{c}^\dagger\right)$

1. Another optical mode: beam-splitter interaction

$$\hat{c} = \hat{b},$$
$$\hat{H}_{\mathrm{BS}} = i\hbar\kappa\left(\hat{a}^\dagger\hat{b} - \hat{a}\hat{b}^\dagger\right).$$

exchanges photons between modes $\hat{a}$ and $\hat{b}$ while conserving the total photon number.

2. An atomic transition: Jaynes-Cummings interaction

$$\hat{c} = \hat{\sigma}_- = |g\rangle\langle e|,$$
$$\hat{c}^\dagger = \hat{\sigma}_+ = |e\rangle\langle g|,$$
$$\hat{H}_{\mathrm{JC}} = i\hbar\kappa\left(\hat{a}^\dagger\hat{\sigma}_- - \hat{a}\hat{\sigma}_+\right).$$

exchanges one excitation between the atom and optical field:

$$|n, e\rangle \longleftrightarrow |n+1, g\rangle$$

3. An atomic population: dephasing interaction

$$\hat{c} = \hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|,$$
$$\hat{H}_{\mathrm{deph}} = i\hbar\kappa\left(\hat{a}^\dagger - \hat{a}\right)\hat{\sigma}_z.$$

produces an atom-dependent displacement of the field

Rotating wave approximation

Suppose $\hat{a}$ and $\hat{c}$ are lowering operators for systems with frequencies ω_a and ω_c. In the interaction picture,

$$\hat{a}(t) = \hat{a}e^{-i\omega_a t}, \qquad \hat{c}(t) = \hat{c}e^{-i\omega_c t}$$

Consider the full quadrature-coupling Hamiltonian

$$\hat{H}_i = -i\hbar\kappa\left(\hat{a} - \hat{a}^\dagger\right)\left(\hat{c} + \hat{c}^\dagger\right)$$

Expanding gives four interaction terms:

$$\hat{H}_i = -i\hbar\kappa\left(\hat{a}\hat{c} + \hat{a}\hat{c}^\dagger - \hat{a}^\dagger\hat{c} - \hat{a}^\dagger\hat{c}^\dagger\right)$$

In the interaction picture,

$$\hat{a}\hat{c} \propto e^{-i(\omega_a+\omega_c)t}, \quad \hat{a}^\dagger\hat{c}^\dagger \propto e^{i(\omega_a+\omega_c)t},$$

$$\hat{a}\hat{c}^\dagger \propto e^{-i(\omega_a-\omega_c)t}, \quad \hat{a}^\dagger\hat{c} \propto e^{i(\omega_a-\omega_c)t}.$$

Near resonance, the rapidly oscillating terms are neglected:

$$\boxed{\hat{H}_i^{\mathrm{RWA}} = i\hbar\kappa\left(\hat{a}^\dagger\hat{c} - \hat{a}\hat{c}^\dagger\right)}$$

Define the detuning as

$$\Delta \equiv \omega_a - \omega_c$$

Exact resonance means

$$\boxed{\Delta = 0} \iff \boxed{\omega_a = \omega_c}$$

From unitary to non-unitary evolution

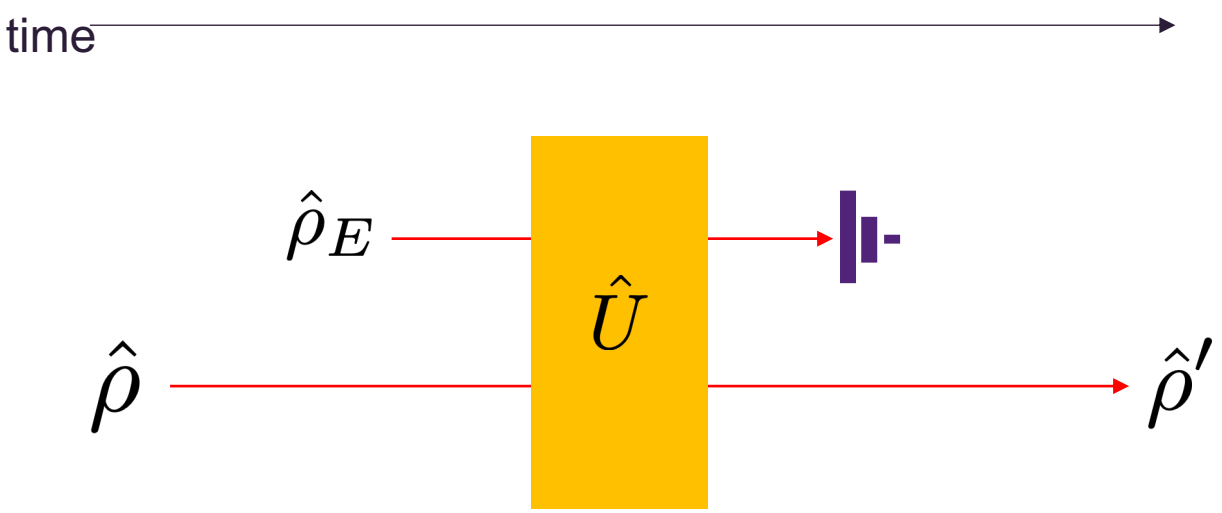
Assume that the system and environment are initially uncorrelated:

$$\hat{\rho}_{SE}(0) = \hat{\rho}_S(0) \otimes \hat{\rho}_E$$

$$\hat{\rho}_S(0) \otimes \hat{\rho}_E \xrightarrow{\hat{U}_{SE}(t)} \hat{\rho}_{SE}(t) \xrightarrow{\mathrm{Tr}_E} \hat{\rho}_S(t)$$

The reduced system state is

$$\hat{\rho}_S(t) = \mathrm{Tr}_E \left[\hat{U}_{SE}(t) \left(\hat{\rho}_S(0) \otimes \hat{\rho}_E \right) \hat{U}_{SE}^{\dagger}(t) \right] \equiv \mathcal{E}_t \left[\hat{\rho}_S(0) \right]$$

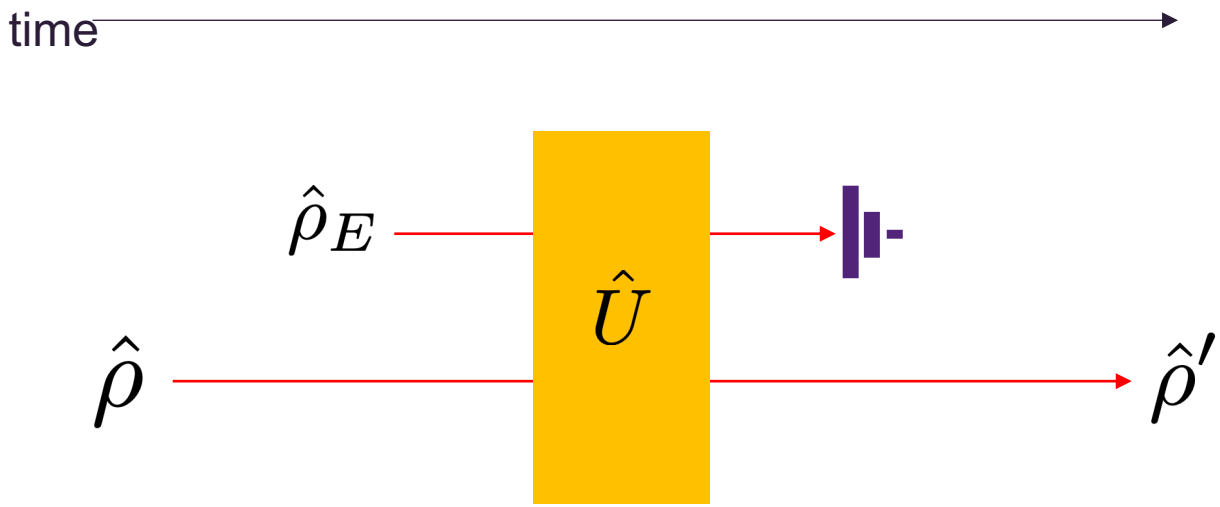


The combined system and environment evolve unitarily.

The environment is then traced out.

The resulting map $\mathcal{E}_t$ is a quantum channel acting on the system.

Why do we have to talk about the joint state of the system and environment?



An interaction can entangle the system and environment. For example,

$$|\Psi_{SE}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_S|0\rangle_E + |1\rangle_S|1\rangle_E\right)$$

The joint state is pure, but the reduced states are mixed:

$$\hat{\rho}_S = \mathrm{Tr}_E\left[|\Psi_{SE}\rangle\langle\Psi_{SE}|\right] = \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right)$$

The combined state may be pure, while the system and environment separately have mixed states.

Modelling optical loss

How do we set up the problem?

Input state on mode A

$$|\psi\rangle_A$$

Combine mode A with environment mode E using beam splitter interaction

$$|\psi\rangle_A \otimes |\phi\rangle_E \rightarrow \hat{U}_B \left[|\psi\rangle_A \otimes |\phi\rangle_E\right]$$

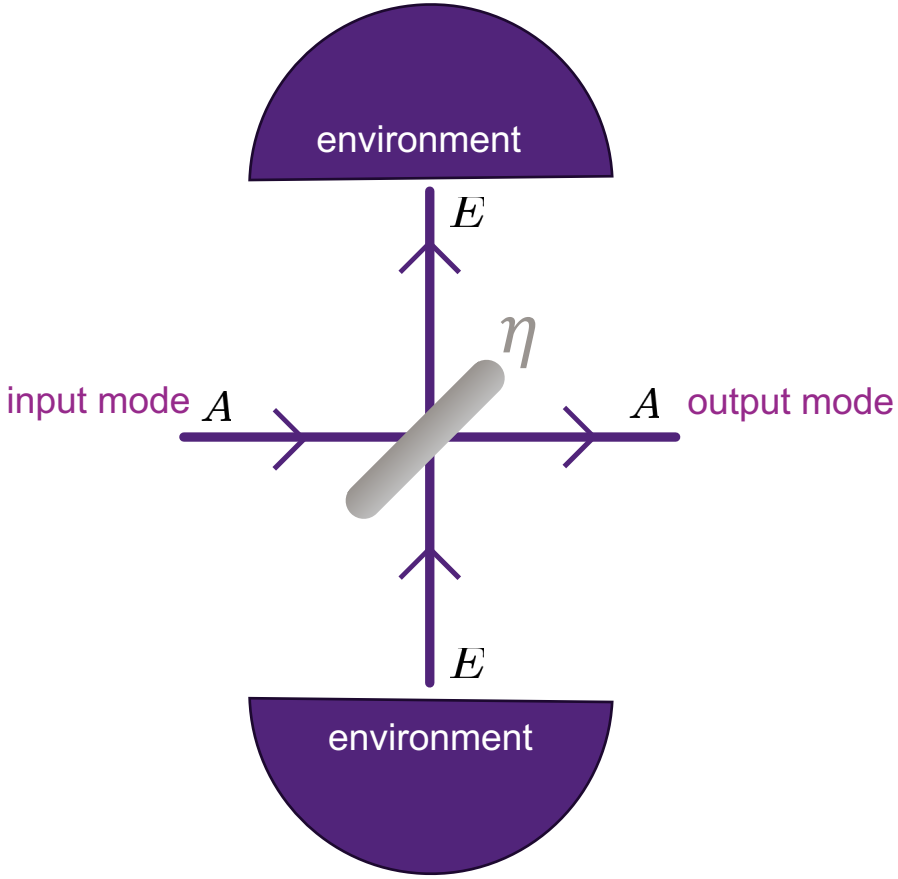
for pure loss, absorption reservoir is the vacuum

$$|\phi\rangle_E = |0\rangle_E$$

To get the reduced density matrix of our system A,
we need to trace over the evironment mode E.

Partial trace over environment mode:

$$\rho_A = \mathrm{Tr}_E \left[\rho_{AE}\right]$$



Modelling optical loss

Single photon input

$$\sqrt{\eta}\,|1\rangle_A\,|0\rangle_E + \sqrt{1-\eta}\,|0\rangle_A\,|1\rangle_E$$

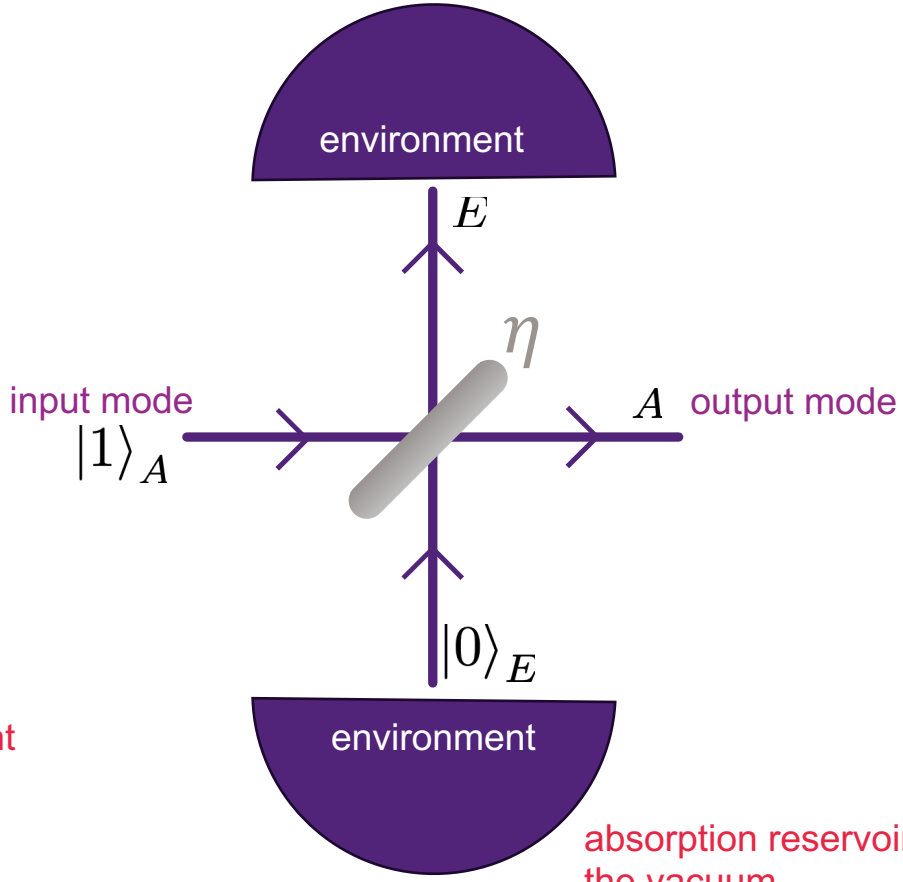
- To get the reduced density matrix of our system A, we need to trace over the evironment mode E.

Trace over environment mode:

$$\begin{aligned}\rho_A &= \mathrm{Tr}_E\left[\rho_{AE}\right] \\ &= \eta\,|1\rangle_A\,\langle 1|_A + (1-\eta)\,|0\rangle_A\,\langle 0|_A\end{aligned}$$

single photon $|1\rangle$ transmitted probability η

single photon in environment probability $1-\eta$



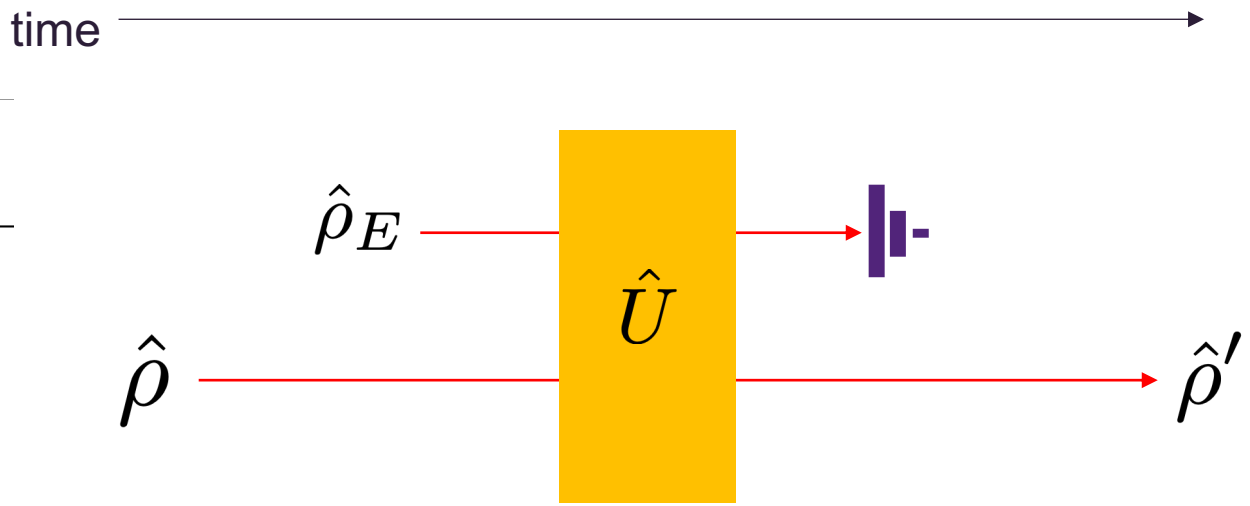
absorption reservoir is the vacuum

- The pure state becomes a mixture of the zero photon term and the one photon term.

Quantum channels: Schrödinger picture

The channel maps the input state to the output state:

$$\hat{\rho}'_S = \mathcal{E}_t\left[\hat{\rho}_S\right] = \mathrm{Tr}_E\left[\hat{U}_{SE}(t)\left(\hat{\rho}_S \otimes \hat{\rho}_E\right)\hat{U}^\dagger_{SE}(t)\right]$$



A physical quantum channel is:

- **Linear**
- **Completely positive**
- **Trace preserving**

Trace preservation requires

$$\mathrm{Tr}\left\{\mathcal{E}_t\left[\hat{\rho}_S\right]\right\} = \mathrm{Tr}\left\{\hat{\rho}_S\right\} = 1$$

In the Schrödinger picture, a quantum channel evolves the state while observables remain fixed.

Quantum channels: Heisenberg picture

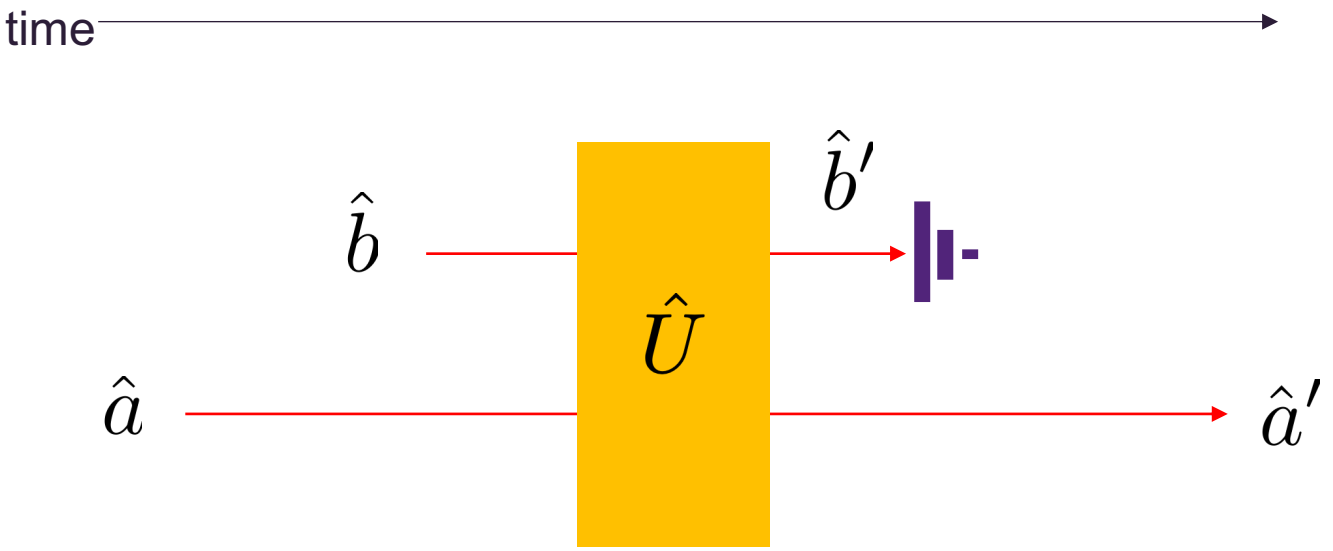
The operators evolve according to

$$\hat{a}' = \hat{U}^\dagger \hat{a} \hat{U}$$

and

$$\hat{b}' = \hat{U}^\dagger \hat{b} \hat{U}$$

In the Heisenberg picture, the input state remains fixed while the operators evolve through the system-environment interaction.



Optical quantum channels as open systems

Aim: use optical quantum channels as our first physical example of an open quantum system.

We want to identify:

- the unitary interaction between the optical system and its environment;
- the initial state of the environment;
- the effective channel produced when the environment is discarded.

Two descriptions

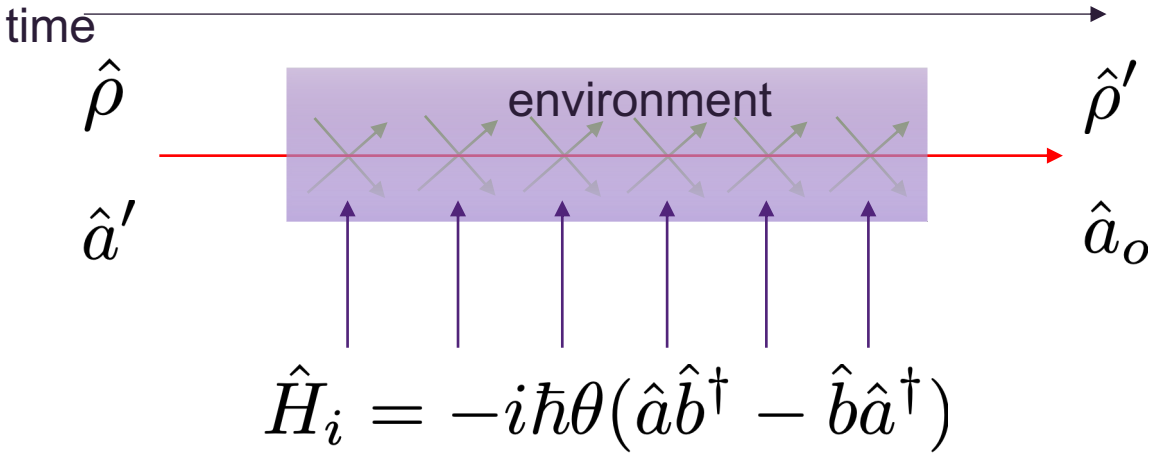
Heisenberg picture: determine how the optical operators transform.

Schrödinger picture: determine how the optical state transforms.

Two physical processes

Physical interaction	Resulting channel
Energy exchange with the environment	Amplitude damping
Environment-induced phase fluctuations	Phase damping

Many weak system-environment interactions



An approximately single-frequency optical mode $\hat{a}$ interacts sequentially with many environmental modes

$$\hat{b}_1, \hat{b}_2, \ldots, \hat{b}_N$$

Each environmental mode is prepared independently in the same state:

$$\hat{\rho}_E^{(N)} = \bigotimes_{j=1}^{N} \hat{\rho}_{E,j}, \qquad \hat{\rho}_{E,j} = \hat{\rho}_E$$

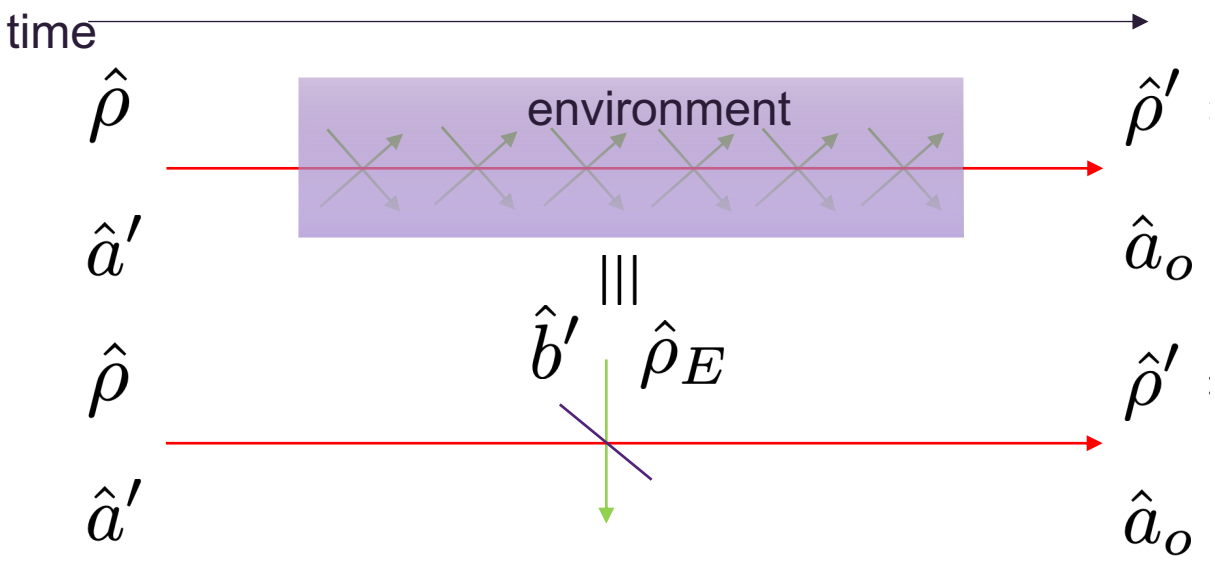
The jth weak interaction is described by

$$\hat{H}_i^{(j)} = i\hbar\kappa_j \left(\hat{a}^\dagger \hat{b}_j - \hat{a}\hat{b}_j^\dagger\right)$$

For an interaction lasting Δt_j, the corresponding beam-splitter angle is small:

$$\theta_j = \kappa_j \Delta t_j \ll 1$$

One effective environmental mode



Many weak interactions with identical thermal modes are equivalent to one stronger interaction with an effective thermal mode.

A sequence of weak beam-splitter interactions can be replaced by one effective beam splitter.

If the jth interaction has transmissivity η_j, the total transmissivity is

$$\eta = \prod_{j=1}^{N} \eta_j$$

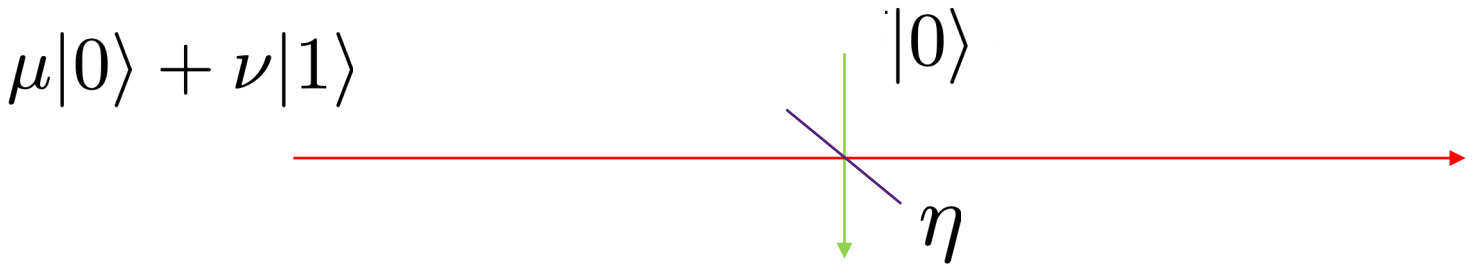
The complete sequence can be written as

$$\hat{a}_{\text{out}} = \sqrt{\eta}\,\hat{a}_{\text{in}} + \sqrt{1-\eta}\,\hat{b}_{\text{eff}}$$

For independent environmental modes prepared in the same thermal state,

$$\hat{\rho}_{b_{\text{eff}}} = \hat{\rho}_E$$

Amplitude-damping channel



- ρ_E: state of the effective environment mode
- η: total intensity transmissivity, with $0 \leq \eta \leq 1$

$$\boxed{\hat{a}' = \sqrt{\eta}\,\hat{a} + \sqrt{1-\eta}\,\hat{b}}$$

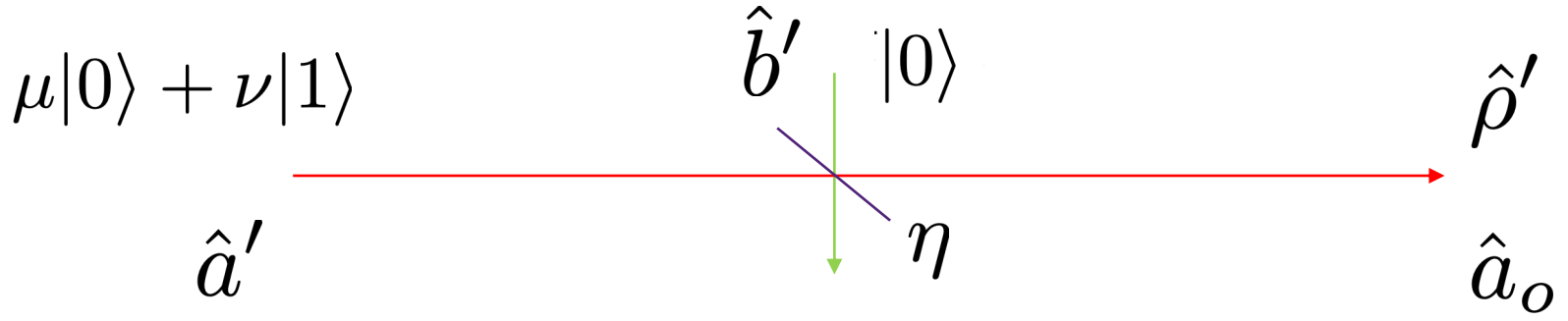
The environmental contribution is required to preserve the bosonic commutation relation:

$$\begin{aligned}\left[\hat{a}', \hat{a}'^{\dagger}\right] &= \eta\left[\hat{a}, \hat{a}^{\dagger}\right] + (1-\eta)\left[\hat{b}, \hat{b}^{\dagger}\right] \\ &= 1.\end{aligned}$$

Channel type:

$$\begin{cases}\text{pure-loss channel}, & \hat{\rho}_E = |0\rangle\langle 0|\,, \\ \text{thermal-loss channel}, & \hat{\rho}_E = \hat{\rho}_{\text{th}}(\bar{n}_E), \quad \bar{n}_E > 0.\end{cases}$$

Pure loss in the Schrödinger picture



Consider a single-rail qubit and a vacuum environment:

$$|\psi\rangle = \mu\,|0\rangle + \nu\,|1\rangle\,, \qquad |\mu|^2 + |\nu|^2 = 1, \qquad \hat{\rho}_E = |0\rangle\langle 0|$$

After the environment is traced out, the output state is

$$\hat{\rho}' = \left[|\mu|^2 + (1-\eta)|\nu|^2\right]|0\rangle\langle 0| + \eta|\nu|^2\,|1\rangle\langle 1| + \sqrt{\eta}\,(\mu\nu^*\,|0\rangle\langle 1| + \mu^*\nu\,|1\rangle\langle 0|)\,.$$

Effect of the channel

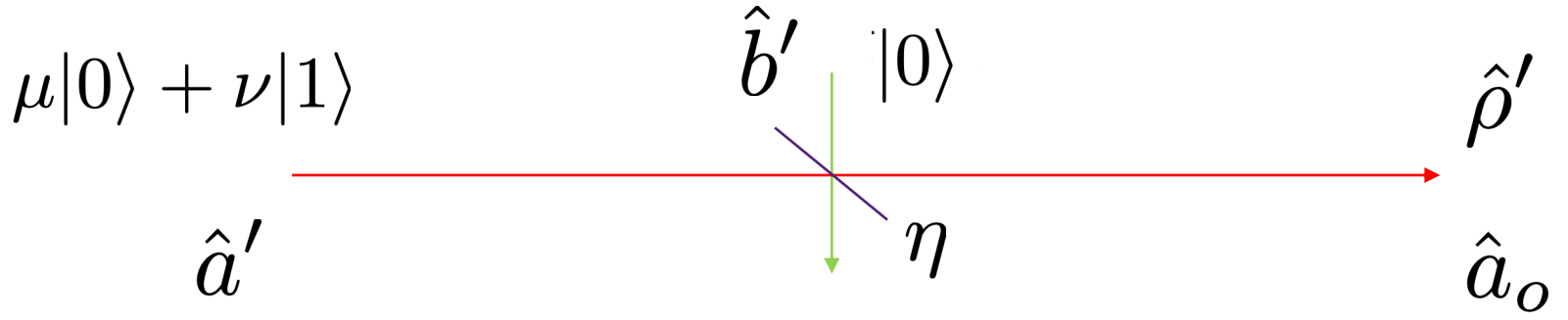
One-photon population:

$$|\nu|^2 \longrightarrow \eta|\nu|^2$$

Lost-photon probability:

$$p_{\text{loss}} = (1-\eta)|\nu|^2$$

Optical quantum noise



The output moments are

$$\langle \hat{n} \rangle_{\text{out}} = \text{Tr}\left(\hat{\rho}'\hat{n}\right)$$

$$\begin{aligned}\langle \hat{n} \rangle_{\text{out}} &= \left(1-\eta|\nu|^2\right)\langle 0|\,\hat{n}\,|0\rangle + \eta|\nu|^2\langle 1|\,\hat{n}\,|1\rangle \\ &= \left(1-\eta|\nu|^2\right)(0) + \eta|\nu|^2(1) \\ &= \eta|\nu|^2.\end{aligned}$$

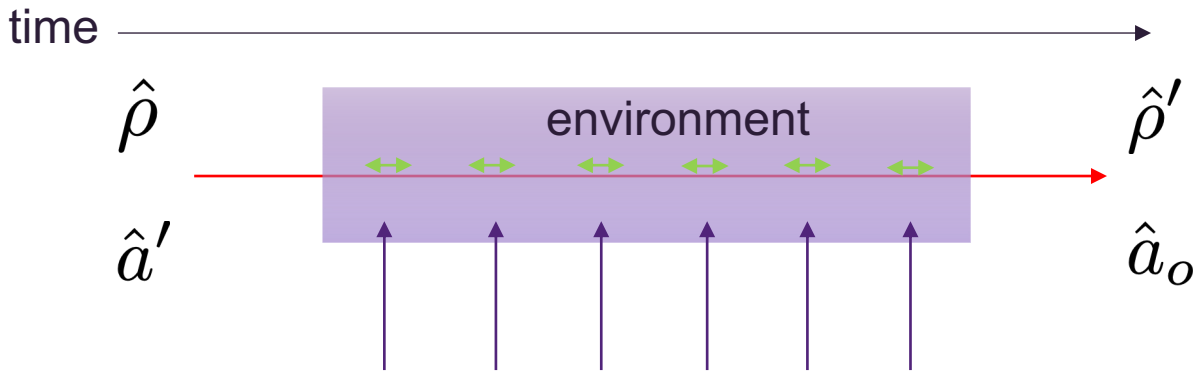
$$\begin{aligned}\langle \hat{n}^2 \rangle_{\text{out}} &= \left(1-\eta|\nu|^2\right)\langle 0|\,\hat{n}^2\,|0\rangle + \eta|\nu|^2\langle 1|\,\hat{n}^2\,|1\rangle \\ &= \left(1-\eta|\nu|^2\right)(0) + \eta|\nu|^2(1) \\ &= \eta|\nu|^2.\end{aligned}$$

$$\langle \hat{n} \rangle = \eta|\nu|^2, \qquad \langle \hat{n}^2 \rangle = \eta|\nu|^2$$

Therefore, the photon-number variance is

$$\boxed{V_n = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = \eta|\nu|^2\left(1-\eta|\nu|^2\right)}$$

Phase damping: coupling to the environment



$$\hat{H}_i = \chi \hat{a}^\dagger \hat{a}(\hat{b} + \hat{b}^\dagger)$$

An approximately single-frequency optical mode interacts weakly with many environmental modes.

Define the system photon-number operator as $\hat{n} = \hat{a}^\dagger \hat{a}$

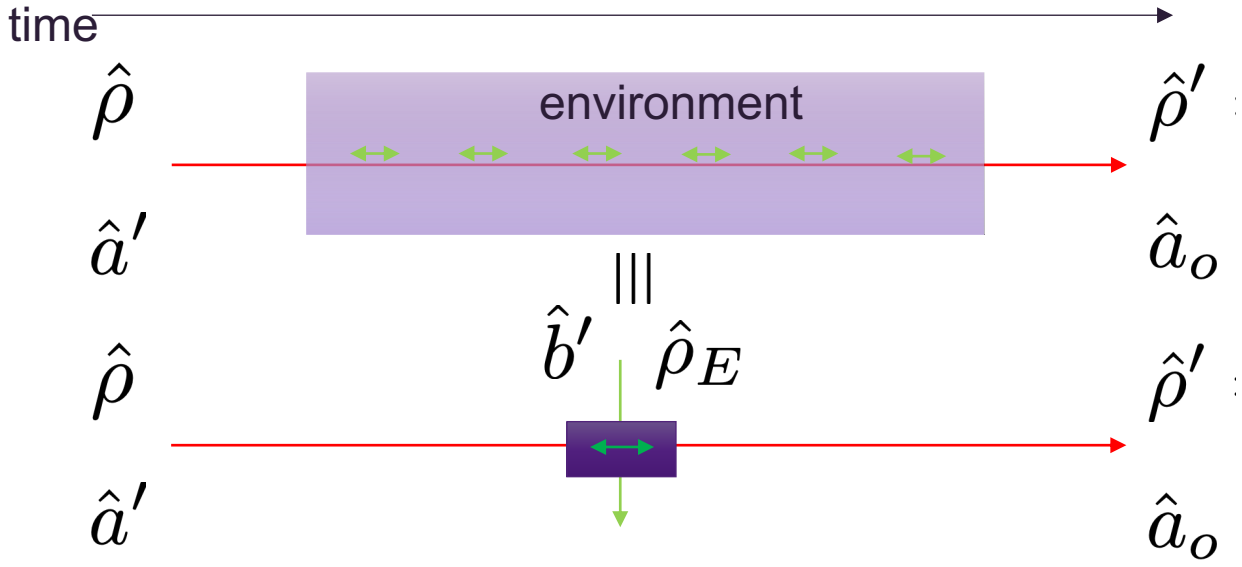
The interaction with the jth environmental mode is

$$\boxed{\hat{H}_i^{(j)} = \hbar\chi_j \hat{a}^\dagger \hat{a}\left(\hat{b}_j + \hat{b}_j^\dagger\right) = \hbar\chi_j \hat{n}\left(\hat{b}_j + \hat{b}_j^\dagger\right)}$$

This interaction does not change the system photon number

$$\boxed{\left[\hat{H}_i^{(j)}, \hat{n}\right] = 0}$$

One effective phase-noise mode



Each weak interaction produces the unitary transformation

$$\hat{U}_j = \exp\left[-i\chi_j \Delta t_j\, \hat{n}\left(\hat{b}_j + \hat{b}_j^\dagger\right)\right]$$

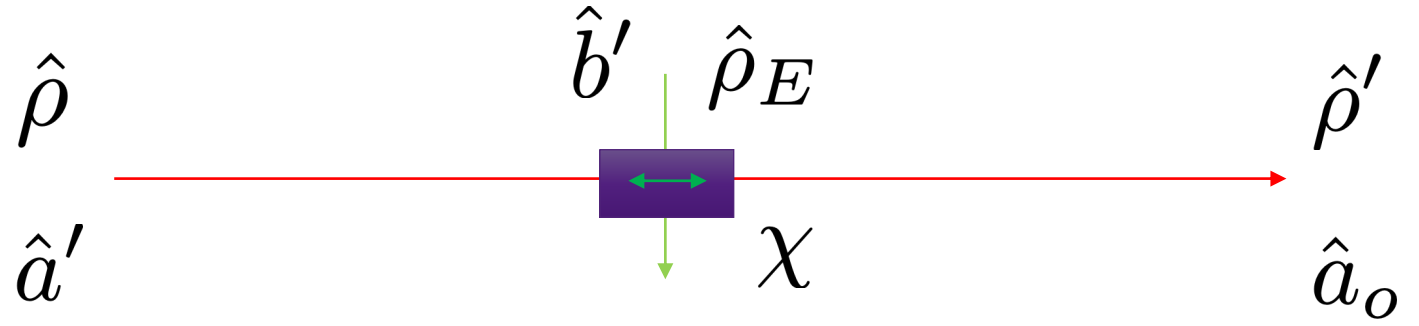
The complete interaction is

$$\prod_{j=1}^{N} \hat{U}_j = \exp\left[-i\hat{n}\sum_{j=1}^{N} \chi_j \Delta t_j \left(\hat{b}_j + \hat{b}_j^\dagger\right)\right]$$

The full effective interaction can be written as

$$\boxed{\hat{U}_{\text{eff}} = \exp\left[-i\chi\hat{n}\left(\hat{b} + \hat{b}^\dagger\right)\right]}$$

Phase-damping channel

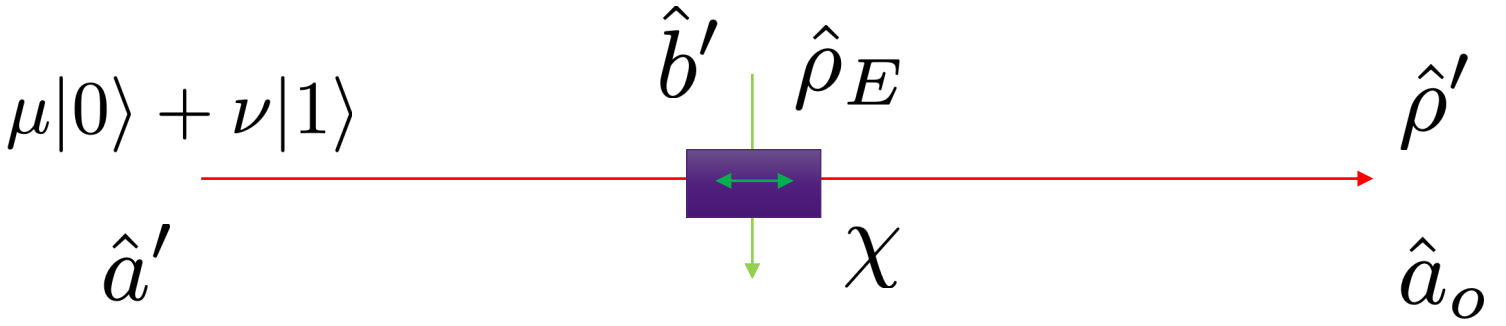


Heisenberg picture

$$\hat{a}' = e^{i\chi(\hat{b}+\hat{b}^\dagger)}\hat{a}$$

completely characterizes channel with phase fluctuation power from uniform environment

Optical quantum decoherence



Schrödinger picture

Consider the input qubit and a vacuum environment:

$$|\psi\rangle = \mu\,|0\rangle + \nu\,|1\rangle\,, \qquad \hat{\rho}_E = |0\rangle\langle 0|$$

After tracing out the environment, the system state becomes

$$\hat{\rho}' = |\mu|^2\,|0\rangle\langle 0| + |\nu|^2\,|1\rangle\langle 1| + e^{-\chi^2/2}\left(\mu\nu^*\,|0\rangle\langle 1| + \mu^*\nu\,|1\rangle\langle 0|\right).$$

$$\rho'_{00} = \rho_{00},$$
$$\rho'_{11} = \rho_{11},$$
$$\rho'_{01} = e^{-\chi^2/2}\rho_{01}.$$

Phase damping leaves the photon-number probabilities unchanged but suppresses the coherence between |0⟩ and |1⟩.

Phase damping suppresses the field quadrature

Photon number is unchanged by phase damping, so we instead measure the phase-sensitive field quadrature:

$$\hat{a} + \hat{a}^\dagger$$

For the input state

$$|\psi\rangle = \mu|0\rangle + \nu|1\rangle$$

the quadrature mean before the channel is

$$\left\langle \hat{a} + \hat{a}^\dagger \right\rangle_{\text{in}} = \mu^*\nu + \nu^*\mu$$

For a vacuum environment, phase damping gives

$$\left\langle \hat{a} + \hat{a}^\dagger \right\rangle_{\text{out}} = \text{Tr}_{S,E}\left[\left(\hat{a}' + \hat{a}'^\dagger\right)\hat{\rho} \otimes \hat{\rho}_E\right]$$

$$= e^{-\chi^2/2}\left(\mu^*\nu + \nu^*\mu\right)$$

Therefore,

$$\boxed{\left\langle \hat{a} + \hat{a}^\dagger \right\rangle_{\text{out}} = e^{-\chi^2/2}\left\langle \hat{a} + \hat{a}^\dagger \right\rangle_{\text{in}}}$$

Summary

- An open-system evolution results from a unitary system-environment interaction followed by tracing over the unobserved environment.

- Many weak environmental interactions can often be represented by a single effective interaction, simplifying the description of an optical channel.

- Amplitude damping exchanges energy and changes photon-number populations, while phase damping preserves photon number but suppresses coherence and phase-sensitive observables.